

Harikumar N

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Education

National Institute of Technology, Rourkela, Integrated MSc Physics 2022 – 2027

CGPA: 6.99/10

Kendriya Vidyalaya , Pattom, Higher Secondary 2021

- Percentage: 96.8
- **Academic Excellence Award:** Received academic excellence award from KVS Sangathan for being in top 1.5 percent of the students who wrote CBSE 2021 BOARD Exam

Experience

National Centre for Radio Astronomy, Master Thesis Intern May 2026 – Present

- Studying Type-I quasar accretion disk variability using ZTF time-series photometry and SDSS survey data. The goal is to connect observed brightness fluctuations to two major physical properties of the accretion disk which contributes to Variability: Reprocessing and thermal fluctuations.

Leiden University, Netherlands, Remote Research Intern May 2025 – Dec 2025

- Validated a structured catalog of absolute spectral-flux calibrators for VLTI/MATISSE and future infrared instruments, enabling precise calibration for high-resolution interferometric observations.
- Conducted comparative analysis of existing Cohen et al. calibrators and new flux models using Python and Pandas. Built visualizations with Matplotlib for spectral comparisons, anomaly detection, and model validation.

Indian Institute of Technology, Hyderabad, Remote Research Intern June 2024 – Feb 2025

- Modelled and analysed 53 archival UV spectra of Seyfert galaxy ESO 141G55 from the IUE satellite. Used JAVELIN CODE to perform reverberation mapping and found its variability.
- Utilized HEASOFT software for data reduction and modeling, employing power-law, Lorentzian models to fit spectral features.
- Quantified flux for both continuum and spectral lines using the cflux model and analyzed

Project

- **Multiwavelength investigation of the infrared dark ring G30.91+0.26:** Worked with Dr Veena V S
- The objective of this project was to conduct a multiwavelength study of the Galactic region centered at G30.91+0.26, with the aim of characterizing its physical and morphological properties. By utilizing data from the Spitzer Space Telescope (infrared) and the SARAO MeerKAT 1.3 GHz Galactic Plane Survey (radio).

Publications

- **Study of UV Line and Continuum Variabilities in the Broadline Seyfert 1 Galaxy ESO 141–G55.** <https://doi.org/10.1002/asna.70093>
Astronomische Nachrichten | Astronomy Journal , 2026.
- **STARSFLUX: an all-sky catalogue of absolute spectro-photometric calibrators from 0.3 to 30 m.** <https://doi.org/10.1093/mnras/stag1478>
Monthly Notices of the Royal Astronomical Society | Astronomy Journal , 2026.

Leadership / Extracurricular

Higher Education Committee, NIT Rourkela, Secretary	2025 - 2026
AstroNITR, Astrophotography Head	2024 - 2025
Analytics and Consulting Club (AnC), Analyst	2023 - 2025

Technical Skills

Programming Languages:Python (NumPy, Matplotlib, Pandas), SQL

Astronomical Data Analysis: *HEASOFT/XSPEC, DS9*

Image Processing and Astrophotography: OpenCV, Matlab, DeepSkyStacker, Registax 6
Software Tools:

- *VS Code, Jupyter Notebook* – IDEs for scripting
- *LaTeX* – Writing research papers, reports, and presentations.

Additional Courses / Certifications

BRICS Astronomy IDIA Data Analytics Training Program	2025
An Observational Overview of Active Galactic Nuclei N. Brandt	2024
Astronomy:Exploring Time and Space, Coursera	2024

Attended Workshops and Conferences

Astronomical Society of India, 2025: Conference at NIT Rourkela	2025
New Frontiers in Astronomy and Astrophysics Workshop	2026
14th Astronomy and Astrophysics School IIST Summer School	2026

Relevant Reading

Active Galactic Nuclei by Bradley Peterson

Acquired a thorough understanding of AGN physics, including accretion processes, emission mechanisms, and the unified model. Learned about observational techniques across the electromagnetic spectrum and the role of AGNs in galaxy evolution.